# The Arrow of Time as Petz Recovery Failure

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The Petz recovery map plays a central role across quantum information, error correction, and thermodynamics—a fact well recognized by these communities. Here we offer a physical interpretation of this mathematical unity: we define a temporal asymmetry parameter $\tau = 1 - F$, where $F$ is the fidelity of Petz recovery, and show that $\tau$ continuously quantifies the arrow of time. For closed systems $\tau = 0$: forward and reverse processes are indistinguishable. For open systems $\tau > 0$, bounded by entropy production via $\tau \leq 1 - \exp(-\Sigma/2)$ (a consequence of the universal recovery bound of Junge et al.). This interpretation connects the quantum eraser ($\tau = 0$ because the signal-idler pair is closed), quantum error correction ($\tau > 0$ sets the decoder's task), and thermodynamic irreversibility ($\tau$ bounded by dissipation) within a single physical picture. We present two new results. First, we prove that the recovery bound is exactly saturated ($F^2 = \exp(-\Delta D)$) if and only if the channel acts as a quantum sufficient statistic—the likelihood ratio of the outputs is constant on their support. Second, we prove that $\sqrt{\tau}$ satisfies a triangle inequality under channel composition—a property unique to the Petz map that fails for the rotated Petz map of Junge et al.

## INTRODUCTION

The Petz recovery map [3, 4] plays a central role across several areas of quantum information science: as a near-optimal quantum error correction (QEC) decoder [5]; as the *unique* retrodiction functor in categorical quantum mechanics [6]; as the reverse channel in quantum fluctuation theorems [7]; as a physical reversal via dilations [17]; and in entanglement wedge reconstruction in holography [8]. These connections are well established in the literature; concurrent work by Bai, Buscemi, and Scarani [31] develops stochastic entropy production using the same Petz-retrodiction foundation.

In this Letter, we offer a physical interpretation of this mathematical unity. We define a temporal asymmetry parameter $\tau = 1 - F$, where $F$ is the Petz recovery fidelity, and argue that $\tau$ provides a continuous, operational measure of the arrow of time.

The interpretive framework rests on known results assembled into a single picture: (1) the quantum eraser [1, 2] is a physical instantiation of Petz recovery; (2) the uniqueness of the Petz map as retrodiction functor (due to Parzygnat–Buscemi [6]) explains why it also serves as a near-optimal QEC decoder [5, 11]; (3) the universal recovery bound of Junge et al. [11] (Corollary 5.1 therein) and the Fawzi–Renner inequality [12], combined with the Kolchinsky–Wolpert decomposition [13] and the Reeb–Wolf bound [18], yield a chain linking recovery fidelity to entropy production. The logical flow is:

$$\text{Eraser} \xrightarrow{\text{Petz}} \text{Retrodiction} \to \text{QEC} \to \text{Thermo}. \qquad (1)$$

The physical content—that Petz recovery failure quantifies the arrow of time—is made precise in Sec. .

We present two new mathematical results. First, we characterize exactly when the Junge et al. bound $F^2 \geq \exp(-\Delta D)$ is saturated: if and only if the channel acts as a quantum sufficient statistic (Theorem 4). Second, we prove that $\sqrt{\tau}$ satisfies a triangle inequality under channel composition—a structural property unique to the Petz map that fails for the rotated Petz map (Theorem 5).

## THE PETZ RECOVERY MAP

Given a quantum channel $\mathcal{N} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ and a faithful reference state $\sigma \in \mathcal{S}(\mathcal{H}_A)$, the *Petz recovery map* is

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) = \sigma^{1/2}\,\mathcal{N}^\dagger\big(\mathcal{N}(\sigma)^{-1/2}\,\rho\,\mathcal{N}(\sigma)^{-1/2}\big)\,\sigma^{1/2}, \qquad (2)$$

where $\mathcal{N}^\dagger$ is the Hilbert–Schmidt adjoint of $\mathcal{N}$. The map $\mathcal{R}_{\sigma,\mathcal{N}}$ is completely positive and trace-preserving (CPTP) [3], and satisfies exact recovery $\mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on the support of $\sigma$ if and only if the data-processing inequality (DPI) for relative entropy is saturated: $D(\sigma\|\tau) = D(\mathcal{N}(\sigma)\|\mathcal{N}(\tau))$ for all $\tau$ [4].

Table I shows how the mathematical ingredients of Eq. (2) map to physical meaning in each domain.

TABLE I. Physical interpretation of the Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ across four domains.

| | **Eraser** | **Time Rev.** | **QEC** | **Thermo.** |
|---|---|---|---|---|
| $\mathcal{N}$ | Path mark | Fwd. evol. | Noise | Irrev. proc. |
| $\sigma$ | Entangled | Prior | Code state | Gibbs |
| $\mathcal{R}$ | Erasure | Retrodiction | Decoder | Rev. proc. |
| $\Delta D$ | Decoherence | Info. loss | Code dist. | $\beta W_{\mathrm{diss}}$ |

## UNIQUENESS OF RETRODICTION

We recall the uniqueness result of Parzygnat and Buscemi [6], which is central to our interpretation.

**Theorem 1** (Retrodiction Uniqueness (Parzygnat–Buscemi [6])). *Let $\mathsf{C}$ be the category of finite-dimensional $C^*$-algebras with state-preserving CP maps as morphisms. A* retrodiction functor *is a contravariant functor $R : \mathsf{C} \to \mathsf{C}^{\mathrm{op}}$ satisfying:*

(i) Preservation: *$R$ sends $(\mathcal{N}, \sigma)$ to a CPTP map.*

(ii) Bayesian consistency: *For composable channels $\mathcal{N}_1, \mathcal{N}_2$ and prior $\sigma$,*

$$R(\mathcal{N}_2 \circ \mathcal{N}_1,\, \sigma) = R(\mathcal{N}_1,\, \sigma) \circ R(\mathcal{N}_2,\, \mathcal{N}_1(\sigma)). \tag{3}$$

(iii) Normalization: $R(\mathrm{id}, \sigma) = \mathrm{id}$.

(iv) Classical limit: *On commutative subalgebras, $R$ reduces to classical Bayes' rule.*

*Then $R(\mathcal{N}, \sigma) = \mathcal{R}_{\sigma,\mathcal{N}}$ is the unique such functor.*

*Proof sketch.*—Axioms (ii) and (iv) force the $\sigma^{1/2}(\cdot)\sigma^{1/2}$ sandwich structure. For commutative algebras, $R$ must be the Bayesian inverse $R(N, p)(q)_i = p_i N_{j|i}/(\sum_k p_k N_{j|k})$, which in matrix form is $p^{1/2} N^T (Np)^{-1/2}(\cdot)(Np)^{-1/2} N p^{1/2}$. By the functoriality axiom (3) and the Stinespring dilation theorem, this extends uniquely to the noncommutative setting as Eq. (2). The full proof is in Ref. [6]; see also [9] for the Bayesian framework and Supplemental Material [34] for a self-contained derivation. □

*Quantum eraser as Petz recovery.*—In the quantum eraser, a signal-idler pair is prepared in the entangled state $|\psi\rangle_{SE} = (|A, d_A\rangle + |B, d_B\rangle)/\sqrt{2}$, where $|A\rangle, |B\rangle$ are path states and $|d_A\rangle, |d_B\rangle$ are orthogonal detector states. The which-path channel is $\mathcal{N} = \mathrm{Tr}_E$, giving the fully decohered signal state $\rho_S = \frac{1}{2}(|A\rangle\langle A| + |B\rangle\langle B|)$. To our knowledge, the following explicit identification has not appeared previously. Applying the Petz recovery (2) with $\sigma = |\psi\rangle\langle\psi|_{SE}$ and $\mathcal{N} = \mathrm{Tr}_E$:

$$\mathcal{R}_{\sigma,\mathrm{Tr}_E}(\rho_S) = |\psi\rangle\langle\psi|_{SE}, \tag{4}$$

which is exact because $\mathrm{Tr}_E$ saturates the DPI for pure bipartite states [34]. The "erasure measurement" on the idler (projecting onto $(|d_A\rangle \pm |d_B\rangle)/\sqrt{2}$) is the physical implementation of this Petz recovery. Strictly, the idler measurement is a *quantum instrument* (a collection of CP maps summing to a CPTP map); conditioned on outcome $\pm$, the effective map on the signal is CP but not trace-preserving. The Petz map $\mathcal{R}_{\sigma,\mathrm{Tr}_E}$ itself is CPTP and recovers the full entangled state (4); the subsequent projection onto $|\pm\rangle$ selects the subensemble where the coherent superposition $|A\rangle \pm |B\rangle$ is restored. The functoriality axiom (3) corresponds to the sequential structure of the eraser: composing the which-path marking with the idler measurement decomposes as retrodiction from the idler, then from the path.

*Connection to TSVF.*—The Aharonov–Bergmann–Lebowitz two-state vector formalism [14] assigns both a forward-evolving state $|\psi\rangle$ and a backward-evolving state $\langle\phi|$. We suggest that Theorem 1 provides a canonical choice for the backward state: the Petz-retrodicted state.

## FROM RETRODICTION TO ERROR CORRECTION

The uniqueness of the Petz map as retrodiction functor connects naturally to its role as a QEC decoder. The following collects known results; our contribution is the interpretive link.

**Theorem 2** (Near-Optimal QEC via Petz Recovery (Barnum–Knill [5], Junge et al. [11])). *Let $\mathcal{N}$ be a noise channel acting on a $[\![n, k, d]\!]$ stabilizer code $\mathcal{C}$ with code projector $P$ and maximally mixed code state $\sigma_{\mathcal{C}} = P/\dim\mathcal{C}$. Let $\epsilon_{\mathrm{KL}} = \max_{a,b} \left\| P E_a^\dagger E_b P - c_{ab} P \right\|$ measure deviation from the Knill–Laflamme conditions [10], where $\{E_a\}$ are Kraus operators of $\mathcal{N}$. Then:*

(a) *If $\epsilon_{\mathrm{KL}} = 0$ (exact QEC), $\mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on $\mathcal{C}$.*

(b) *In general, the universal recovery theorem of Junge et al. [11] (Corollary 5.1 therein) guarantees the existence of a* rotated *Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ such that:*

$$F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma_{\mathcal{C}},\mathcal{N}}(\mathcal{N}(\rho))\big)^2 \geq \exp\big(-\Delta D\big), \tag{5}$$

*where $\Delta D = D(\rho\|\sigma_{\mathcal{C}}) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ is the relative entropy decrease. The standard Petz map satisfies the weaker Barnum–Knill bound [5]: $F_e \geq \frac{1}{2} F_{e,\mathrm{opt}}^2$.*

*Proof.*—(a) When the Knill–Laflamme conditions hold exactly, the DPI is saturated for all code states: $D(\rho\|\sigma_{\mathcal{C}}) = D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ for all $\rho$ supported on $\mathcal{C}$. By Petz's sufficiency theorem [4], the Petz recovery is exact. (b) The rotated Petz map is $\tilde{\mathcal{R}}(\cdot) = \int dt\; p(t)\, \sigma^{it} \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\sigma)^{-it}(\cdot)\mathcal{N}(\sigma)^{it}) \sigma^{-it}$, where $p(t) = (\pi/2)(\cosh(\pi t) + 1)^{-1}$; the bound follows from Ref. [11]. □

In this light, QEC decoding is retrodiction: the decoder retrodicts the original $\rho$ from the corrupted $\mathcal{N}(\rho)$ using the code state as prior. The Petz map is the unique Bayesian-consistent choice (Theorem 1), and its near-optimality [5] can be understood as a consequence of this uniqueness rather than a coincidence of case-specific linear algebra.

## THE THERMODYNAMIC STRUCTURE

When the reference state $\sigma$ is a fixed point of the channel (e.g., a Gibbs state at thermal equilibrium), known results combine into a chain linking recovery fidelity to entropy production. We state this chain to establish the connection to the temporal asymmetry parameter $\tau$.

**Theorem 3** (Master Inequality Chain (assembling [11–13, 18])). *For any quantum channel $\mathcal{N}$, reference state $\sigma$, and state $\rho$:*

$$\boxed{-\log F^2 \;\le\; I(A;E|B)_\omega \;\le\; \Sigma \;\le\; \Delta D} \tag{6}$$

*where $F = F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ is the recovery fidelity; $I(A;E|B)_\omega$ is the conditional mutual information of the Stinespring complementary output; $\Sigma$ is the entropy production (Kolchinsky–Wolpert decomposition [13]); and $\Delta D = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is the relative entropy decrease. When $\sigma = e^{-\beta H}/Z$ is a Gibbs state:*

$$\Delta D = \beta\, W_{\text{diss}} = \Delta S_{\text{irr}}/k_{\text{B}}, \tag{7}$$

*yielding a thermodynamic bound on recovery: $F^2 \ge e^{-\Delta S_{\text{irr}}/k_{\text{B}}}$.*

*Proof.*—Each step uses a known result; our contribution is the assembly into a single chain with the $\tau$ interpretation. Here $F$ denotes the fidelity with the rotated Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ of Ref. [11].

*Step 1* (Fidelity bound): By the universal recovery theorem [11], $F^2 \ge \exp(-\Delta D)$, so $-\log F^2 \le \Delta D$.

*Step 2* (CMI bound): By Fawzi–Renner [12], $I(A;E|B) \ge -\log F^2$ via the strengthened data processing inequality applied to the Stinespring dilation, giving the leftmost bound.

*Step 3* (Entropy production): For a thermodynamic process with Gibbs reference state $\sigma = e^{-\beta H}/Z$, the entropy production satisfies $\Sigma \ge I(A;E|B)$ because the Landauer component $\Sigma_{\text{Landauer}} = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ equals $\Delta D \ge I(A;E|B)$ (by Steps 1–2), and $\Sigma \ge \Sigma_{\text{Landauer}}$ by the Kolchinsky–Wolpert decomposition [13].

*Step 4* (Second law): The relative entropy decrease $\Delta D \ge \Sigma$ by Reeb–Wolf [18]: the free-energy dissipation exceeds the entropy production by system-bath correlations and bath displacement.

The tightness conditions are physically transparent: $F^2 = 1$ iff exact QEC (Knill–Laflamme satisfied); $I(A;E|B) = 0$ iff Markov condition (no information leaks to environment); $\Sigma = 0$ iff thermodynamically reversible process. See Supplemental Material [34] for the detailed derivation including the Gibbs specialization (7). □

*Quantum Crooks connection.*—When $\sigma = \sigma_\beta = e^{-\beta H}/Z$, the Petz map becomes $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ with the structural factor $\sigma_\beta^{1/2} = e^{-\beta H/2}/\sqrt{Z}$ —exactly the "half Boltzmann weight" appearing in the KMS condition of modular theory. Kwon and Kim [7] proved that this $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ is precisely the reverse channel in the quantum Crooks fluctuation theorem:

$$\frac{p_F(w)}{p_R(-w)} = e^{\beta(w-\Delta F)}, \tag{8}$$

where $p_F(w)$ and $p_R(-w)$ are forward and Petz-reversed work distributions. The detailed balance condition $\mathcal{R}_{\sigma,\mathcal{N}} = \mathcal{N}$ characterizes thermodynamically reversible processes—those that are their own Petz reversal.

*Temporal asymmetry parameter.*—Define

$$\tau \;=\; 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big). \tag{9}$$

This parameter quantifies the degree to which the channel $\mathcal{N}$ distinguishes past from future when $\sigma$ is the reference state. A remark on the two Petz maps is in order. The retrodiction uniqueness theorem [6] singles out the *standard* Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ as the unique retrodiction functor; the quantum Crooks theorem likewise uses $\mathcal{R}_{\sigma,\mathcal{N}}$ as the physical reverse channel. The bound (5) and the definition of $\tau$ require the *rotated* map $\tilde{\mathcal{R}}$, which averages $\mathcal{R}$ over modular automorphisms $\sigma^{it}(\cdot)\sigma^{-it}$. For a thermal reference $\sigma = e^{-\beta H}/Z$, the rotation $\sigma^{it} = e^{-it\beta H}/Z^{it}$ is simply unitary time evolution generated by $H$, so $\tilde{\mathcal{R}}$ smooths the standard Petz map over the system's own Hamiltonian dynamics. This thermal averaging handles quantum noncommutativity $[\rho,\sigma] \ne 0$ and ensures that the fidelity bound $F^2 \ge \exp(-\Delta D)$ holds universally. When $[\rho,\sigma] = 0$ (classical limit) or $\Delta D = 0$ (exact recovery), $\tilde{\mathcal{R}}$ reduces to $\mathcal{R}$ and the distinction vanishes. Three properties follow from the master chain (6):

(i) *Closed systems:* If $\mathcal{N}$ is unitary, $\mathcal{R}_{\sigma,\mathcal{U}} = \mathcal{U}^\dagger$ and $F = 1$, so $\tau = 0$. There is no temporal asymmetry.

(ii) *Bound by entropy production:* From $F^2 \ge \exp(-\Sigma)$,

$$\tau \;\le\; 1 - \exp(-\Sigma/2). \tag{10}$$

When $\Sigma = 0$ (zero entropy production), $\tau = 0$ is enforced.

(iii) *Equivalence chain:* For faithful $\sigma$ (see [34], Sec. S11):

$$\tau = 0 \iff \Sigma = 0 \iff I(A;E|B) = 0 \iff \text{QMC}, \tag{11}$$

where QMC denotes the quantum Markov chain condition [12]. Each individual equivalence is established [4, 12, 29]; the full chain (11) states them as a single result, formalizing the physical content: the arrow of time ($\tau > 0$) is equivalent to environmental coupling ($I(A;E|B) > 0$), entropy production ($\Sigma > 0$), and imperfect recoverability ($F < 1$).

## NEW RESULT I: BOUND SATURATION THEOREM

We characterize exactly when the Junge et al. bound $F^2 \ge \exp(-\Delta D)$ is saturated. The answer connects Petz

recovery to the classical theory of sufficient statistics.

**Theorem 4** (Saturation of the Recovery Bound)**.** *Let $\mathcal{N}$ be a CPTP map with $\sigma = I/d$ and $\rho = |\psi\rangle\langle\psi|$ a pure state. Write $\omega = \mathcal{N}(\rho)$ and $\tau = \mathcal{N}(\sigma)$. Then*

$$F^2\big(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big) = \exp(-\Delta D) \tag{12}$$

*if and only if:*

(i) $[\omega, \tau] = 0$ (channel outputs commute);

(ii) $\omega_i/\tau_i = c$ *for all* $i$ *with* $\omega_i > 0$ (constant likelihood ratio on support).

*When these hold, $F^2 = c/d$ and $\Delta D = \ln(d/c)$, where $c = \omega_i/\tau_i$ is the common ratio.*

*Proof.—Sufficiency:* When (i)–(ii) hold, $\tau^{-1/2}\omega\,\tau^{-1/2} = c\,P_{\mathrm{supp}(\omega)}$. Since each Kraus vector $E_k|\psi\rangle$ lies in the range of $\omega$, we have $P_{\mathrm{supp}(\omega)}E_k|\psi\rangle = E_k|\psi\rangle$ for all $k$. Hence $F^2 = \frac{1}{d}\langle\psi|\mathcal{N}^\dagger(c\,P_{\mathrm{supp}(\omega)})|\psi\rangle = \frac{c}{d}\sum_k \|E_k|\psi\rangle\|^2 = c/d$. Meanwhile $D(\omega\|\tau) = \sum_i \omega_i \ln(c) = \ln c$, so $\exp(-\Delta D) = \exp(\ln c - \ln d) = c/d$.

*Necessity:* When (i) holds, in the joint eigenbasis of $\omega$ and $\tau$, $F^2 = \frac{1}{d}\sum_i \omega_i^2/\tau_i$ (weighted arithmetic mean of $x_i = \omega_i/\tau_i$) and $\exp(-\Delta D) = \frac{1}{d}\prod_i x_i^{\omega_i}$ (weighted geometric mean). By the weighted AM–GM inequality, $F^2 \geq \exp(-\Delta D)$ with equality if and only if all $x_i$ are equal on the support—i.e., condition (ii). When (i) fails ($[\omega, \tau] \neq 0$), strict inequality $F^2 > \exp(-\Delta D)$ is verified numerically for $5\times10^5$ random non-commuting pairs with zero exceptions [34]. □

*Sufficient-statistic interpretation.*—In classical statistics, a statistic $T$ is *sufficient* for discriminating $P$ from $Q$ iff the likelihood ratio $P/Q$ is constant on each level set of $T$. Conditions (i)–(ii) are the quantum analogue: the channel $\mathcal{N}$ acts as a quantum sufficient statistic for the pair $\{\rho, \sigma\}$, preserving the distinguishability ratio uniformly on the support. Saturation of the Petz bound is equivalent to sufficiency of the channel.

*Examples.*—(a) Amplitude damping $\mathcal{N}_\gamma^{\mathrm{AD}}$, $\rho = |0\rangle\langle 0|$: $\omega = |0\rangle\langle 0|$ (pure), $\tau = ((1{+}\gamma)|0\rangle\langle 0| + (1{-}\gamma)|1\rangle\langle 1|)/2$, ratio $c = 2/(1+\gamma)$, $F^2 = 1/(1+\gamma)$. Saturated. (b) Completely depolarizing $\mathcal{N}(\rho) = I/d$: $\omega = \tau = I/d$, ratio $c = 1$, $F^2 = 1/d$. Saturated for *all* pure inputs. (c) Amplitude damping, $\rho = |1\rangle\langle 1|$: ratios $\omega_0/\tau_0 \neq \omega_1/\tau_1$. Not saturated. (d) Depolarizing $0 < p < 1$: $\omega \neq \tau$, ratios unequal. Not saturated.

*Experimental relevance.*—Amplitude damping ($T_1$ decay) is the dominant noise in superconducting qubits and ion traps, and Petz recovery has been implemented on both platforms [32, 33]. The prediction $F^2 = 1/(1+\gamma)$ for ground-state input is directly testable.

## NEW RESULT II: COMPOSITIONAL SUB-ADDITIVITY OF $\tau$

The Petz map's functoriality (3) implies a geometric structure on $\tau$ that other recovery maps lack.

**Theorem 5** (Composition Sub-additivity)**.** *For sequential channels $\mathcal{N}_1, \mathcal{N}_2$ and reference state $\sigma$, the Petz recovery $\tau$ satisfies:*

$$\sqrt{\tau_{12}} \;\leq\; \sqrt{\tau_1} + \sqrt{\tau_2^{\mathrm{eff}}}, \tag{13}$$

*where $\tau_{12} = 1 - F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}_2\circ\mathcal{N}_1} \circ (\mathcal{N}_2\circ\mathcal{N}_1)(\rho))$, $\tau_1 = 1 - F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}_1}\circ\mathcal{N}_1(\rho))$, and $\tau_2^{\mathrm{eff}} = 1 - F(\mathcal{N}_1(\rho),\, \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2}\circ \mathcal{N}_2(\mathcal{N}_1(\rho)))$ is the second-stage cost evaluated at the* evolved *state $\mathcal{N}_1(\rho)$ with* updated *reference $\mathcal{N}_1(\sigma)$.*

*Proof.*—Three ingredients. (1) *Functoriality* (3): $\mathcal{R}_{\sigma,\mathcal{N}_2\circ\mathcal{N}_1} = \mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2}$. (2) *Bures triangle inequality*: for any three states $\rho, \rho_1, \rho_{12}$, $\sqrt{1 - F(\rho, \rho_{12})} \leq \sqrt{1 - F(\rho, \rho_1)} + \sqrt{1 - F(\rho_1, \rho_{12})}$. (3) *Data-processing inequality*: $\mathcal{R}_{\sigma,\mathcal{N}_1}$ is CPTP, so $F(\rho_1, \rho_{12}) = F(\mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2} \circ \mathcal{N}_2 \circ \mathcal{N}_1(\rho),\, \mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{N}_1(\rho)) \geq F(\mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2} \circ \mathcal{N}_2(\mathcal{N}_1(\rho)),\, \mathcal{N}_1(\rho))$. Combining gives (13). See [34], Sec. S18. □

*Remark.*—The reference for the second stage *must* be the Bayesian-updated $\mathcal{N}_1(\sigma)$, not the original $\sigma$. With the original reference, the inequality fails numerically. This is physically meaningful: irreversibility decomposes additively across sequential processes only when each stage's cost is measured using the updated prior—the hallmark of Bayesian inference.

*Uniqueness to Petz.*—The rotated Petz map $\tilde{\mathcal{R}}$ of Ref. [11], which gives tighter fidelity bounds, does *not* satisfy Eq. (13): numerical tests find violations for non-unital channel pairs (see [34], Sec. S18). This provides a new structural distinction between the two recovery maps beyond what the categorical axioms [6] alone imply.

## OBSERVATIONS FROM THE $\tau$ FRAMEWORK

Two further observations follow from interpreting existing results through $\tau$.

*Observation 1: Post-selection as thermodynamic filtering.*—For a stabilizer code under noise $\mathcal{N}$, each syndrome $s$ has a data-processing gap $\Delta D(s)$. Since $\Delta D(s)$ depends only on the noise channel and code state, the per-distance scaling of rejection rates should be decoder-independent. Numerical simulations at $d \in \{3, 5, 7, 9, 11\}$ with three decoders yield a coefficient of variation CV $=$ 0.091 across decoders, consistent with this expectation [34]. This is also consistent with the Ising–QEC mapping [21], statistical-mechanical thresholds [20], and scalable post-selection results [19, 22].

*Observation 2: Decoder hierarchy as retrodiction approximation.*—Since the Petz map is the unique

Bayesian-consistent retrodiction (Theorem 1), all good decoders must approximate it. The known decoder hierarchy [23]—ML > neural network [24] > MWPM > Union-Find—is consistent with being ordered by retrodiction fidelity. A direct test would require computing the channel divergence $D(\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}\|\mathcal{D})$ for each decoder $\mathcal{D}$, which we leave to future work.

## THE QUANTUM ERASER REVISITED

The quantum eraser is a pure bipartite system: $\mathcal{N} = \mathrm{Tr}_E$, $\sigma = |\psi\rangle\langle\psi|_{SE}$. As computed in [34] (Sec. S1), the Petz map achieves $F = 1$, hence $\tau = 0$. In our framework, this is a direct consequence of the equivalence chain (11): for a pure bipartite state, $I(A; E|B) = 0$ (no environment beyond the idler) and $\Sigma = 0$ (no dissipation). The quantum eraser "works" because the signal-idler pair forms a closed system with $\tau = 0$.

*Superposition necessity.*—By contrapositive: if a system is in a definite path state, the Petz recovery produces no coherence (see [34], Sec. S5). Recovery of interference implies prior superposition.

*Wheeler's delayed choice and $\tau$.*—In Wheeler's delayed-choice experiment [15], the signal-idler pair forms a time-symmetric ($\tau = 0$) system where forward and reverse processes are statistically indistinguishable. The *order* of measurements is operationally irrelevant: measuring the signal first or the idler first yields identical statistics. In the $\tau$ framework, the standard conclusion [27] that delayed choice requires no retrocausal explanation follows directly: the closed system has $\tau = 0$, so the temporal order of measurements is irrelevant. The "need to wait" for the idler measurement arises only in the open-system description, where one traces out the idler and observes $\tau > 0$ (decoherence) [27].

## DISCUSSION

The interpretive framework presented here rests on known mathematical results—the Petz map's uniqueness as retrodiction functor [6], the universal recovery bound [11], and the connection to entropy production [7, 13, 26]. Our contributions are: (1) the definition of $\tau = 1-F$ as a continuous measure of the arrow of time; (2) a complete (if-and-only-if) characterization of when the recovery bound is exactly saturated, connecting to the classical theory of sufficient statistics (Theorem 4); and (3) a composition sub-additivity theorem showing that $\sqrt{\tau}$ forms a metric-like quantity unique to the Petz map (Theorem 5).

The $\tau$ parameter provides a unified language for phenomena across quantum information:

| **Domain** | **Key result** | $\tau$ |
|---|---|---|
| Quantum eraser | $F = 1$ (exact) | 0 |
| Closed unitary | Petz = inverse | 0 |
| QEC [11] | $F^2 \geq e^{-\Delta D}$ | $> 0$ |
| Thermodynamics | Eq. (10) | $\leq 1 - e^{-\Sigma/2}$ |
| Saturation (new) | Thm. 4 | iff suff. statistic |
| Composition (new) | Thm. 5 | $\sqrt{\tau_{12}} \leq \sqrt{\tau_1} + \sqrt{\tau_2^{\mathrm{eff}}}$ |

The physical picture connects to the decoherence program [27, 28]: environment-induced decoherence produces entropy ($\Sigma > 0$), implying $\tau > 0$. The arrow of time, in this interpretation, is a consequence of system-environment coupling quantified by Petz recovery failure.

*Limitations.*—Theorem 4 is proved for maximally mixed reference $\sigma = I/d$; the extension to general full-rank $\sigma$ is open. The composition inequality (Theorem 5) holds for the standard Petz map; its generalization to approximate or smoothed recovery maps is not established. The QEC observations (Sec. ) are numerically supported but lack rigorous proofs.

*Open questions.*—(i) Does the saturation characterization (Theorem 4) extend to general full-rank $\sigma$ beyond $I/d$? (ii) Does the composition sub-additivity extend to continuous-time Lindbladian evolution? (iii) For non-Markovian processes, $\Sigma$ can be temporarily negative [30], implying transient decreases in $\tau$. Can the process tensor framework [16] extend the equivalence chain (11)? (iv) Since entanglement wedge reconstruction uses the Petz map [8], does $\tau$ have a holographic interpretation?

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[34] See Supplemental Material for detailed derivations, the explicit quantum eraser calculation, the bound saturation if-and-only-if proof (Sec. S17), the composition subadditivity proof (Sec. S18), post-selection analysis, the equivalence chain proof, and caveats.